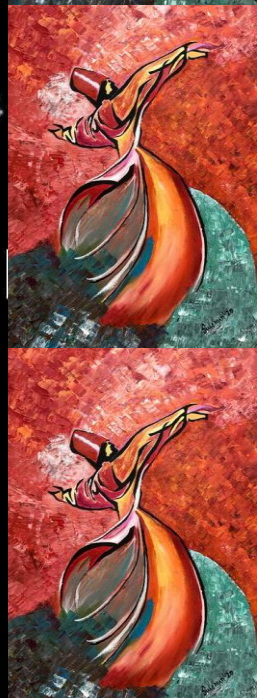




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



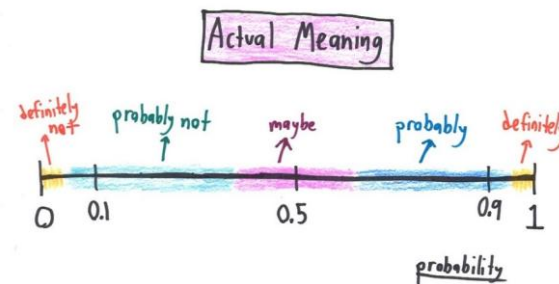


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 16 – Lecture No 31 - April 29, 2026





*Thank you &
Good Luck
The Last Lecture*

وہ بات سارے فسانے میں جس کا ذکر نہ تھا
وہ بات ان کو بہت ناگوار گزری ہے

Announcements!

1. End-term Exam - Monday, May 4, 2026 (8:30a to 10:30a) @ Tariq Rafi, E-220

1. Cheat Sheet - double-sided A-4 (HAND WRITTEN ONLY notes)
2. Fall 2025 end-term uploaded
3. 5 questions - 2 hrs

Agenda for today

1. End-Term Exam
2. Unit 6 - Derived Distributions (if only time had permitted)
- ➡ 3. Quiz No 10 - April 29, 2026 (Wednesday) - Bayes Rule

Instructions, please!

○ End-term Exam – Monday, May 4, 2026 (8:30a to 10:30a) @ Tariq Rafi, E-220

1. Arrive 10 minutes before the start of the exam.
2. **Mobile phones. Smart watches and other electronic devices must be turned off and put away before the start of the exam.**
3. **You are allowed to use a scientific calculator (Not the one on your smart phone. Sharing of calculators is not allowed as well).**
4. No food or drink, other than water, is allowed in the examination room.
5. You are not allowed to communicate with other candidates.
6. You are not allowed to leave your seat unless granted permission by an invigilator.
7. Assume where necessary and justify your assumptions.
8. You are required to fill all answers in the provided answer booklet.
9. **You are allowed to bring a cheat sheet. It must be a handwritten A-4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are not allowed.**
10. The standard normal table (if needed) will be provided

Example Bayes Rule – Continuous – Discrete RVs

Example pertains to inferring a continuous random variable Q , from a discrete observation of a random variable X . Let Q be a continuous random variable with PDF as follows:

$$f_Q(q) = \begin{cases} 6q(1 - q), & \text{if } 0 \leq q \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Q represents the probability of a success of a Bernoulli random variable X , i.e., $P(X = 1 | Q = q) = q$ & $P(X = 0 | Q = q) = (1 - q)$. Find $f_{Q|X}(q|x)$ for $x \in \{0, 1\}$, for all possible values of q . You may use following version of Bayes' rule:

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$\text{Given } f_{\theta|X}(u|x) = \frac{f_{\theta}(u) \cdot P_{X|\theta}(x|u)}{P_X(x)}$$

$$\text{where } P_X(x) = \int_0^1 f_{\theta}(u) \cdot P_{X|\theta}(x|u) du \quad \text{for } x=0,1 \text{ \& } u \in [0,1]$$

for $x=0$

$$f_{\theta|X}(u|0) = \frac{6u(1-u) \cdot (1-u)}{\int_0^1 6u(1-u)(1-u) du} \quad ; \quad P_{X|\theta}(0|u) = (1-u)$$

$$= \frac{6u(1-u)(1-u)}{6 \int_0^1 u + u^3 - 2u^2 du}$$

$$= \frac{6u(1-u)(1-u)}{6 \left[u^2/2 + u^4/4 - 2u^3/3 \right]_0^1} = \frac{6u(1-u)(1-u)}{1/2}$$

$$= 12u(1-u)^2$$

$$0 \leq u \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

for $x=1$ and $q \in [0,1]$

$$f_{Q|X}(q|1) = \frac{6q(1-q) \cdot q}{1/2}$$

as $P_{X|Q}(1|q) = q$

$P_X(0) = 1/2 \Rightarrow$ ^{implies} $P_X(1) = 1/2$

$$f_{Q|X}(q|1) = 12q^2(1-q)$$

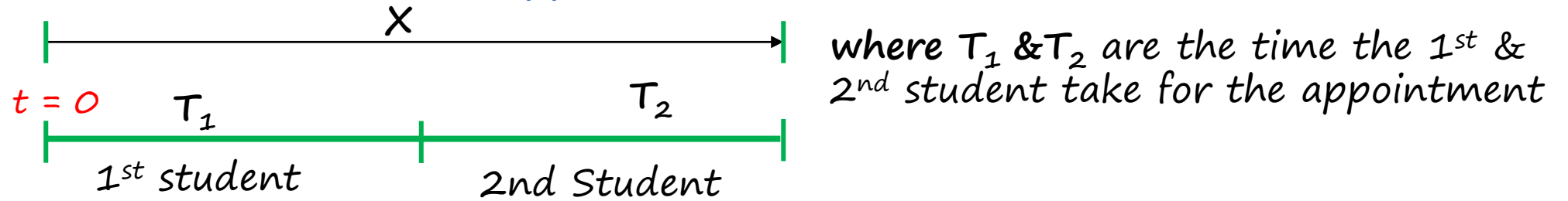
$$0 \leq q \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

Home Assignment

Example: Absent minded Professor – 2 appointments w/ two students at the same time



$$T_1 \text{ \& } T_2 \sim \exp(\lambda), \quad P(T_1 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0 \quad P(T_2 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0$$

$$E[T_1] = E[T_2] = 1/\lambda$$

Given that 2nd student arrives exactly after 5 minutes of the arrival of the 1st student

Comment on the $E[X]$ – where X is the total duration of the two appointments & for both students $\lambda = 1/30$ mins

$$\text{Case 1 – condition } T_1 < 5 \text{ mins} \quad E[X \mid T_1 < 5] = 5 + E[T_2] = 35 \text{ mins}$$

$$\text{Case 2 – condition } T_1 > 5 \text{ mins} \quad E[X \mid T_1 > 5] = 5 + E[T_1] + E[T_2] = 5 + 30 + 30 = 65 \text{ mins}$$

$$E[X] = P(T_1 < t) \cdot E[X \mid T_1 < 5] + P(T_1 > t) \cdot E[X \mid T_1 > 5]$$

Question 2 - [6 points]: A wire has length X , an exponential random variable with parameter $\lambda = 1/\pi$ cm. The wire is cut to make rings of diameter 1 cm. Find the PMF $P_N(n)$, the probability for the number of complete rings N produced by each length of wire.

Hint – use CDF $F_X(x)$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- Continuous random variables:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Question 2 - [6 points]: A wire has length X , an exponential random variable with parameter $\lambda = 1/\pi$ cm. The wire is cut to make rings of diameter 1 cm. Find the PMF $P_N(n)$, the probability for the number of complete rings N produced by each length of wire.

Hint – use CDF $F_X(x)$

$$P(N=0) = F_X(\pi)$$

$$P(N=1) = F_X(2\pi) - F_X(\pi)$$

$\vdots$

$$P(N=n) = F_X((n+1)\pi) - F_X(n\pi)$$

$$= 1 - e^{-\lambda(n+1)\pi} - (1 - e^{-\lambda n\pi}) \quad \text{--- 2}$$

$$= 1 - e^{-\lambda(n+1)\pi} + e^{-\lambda n\pi}$$

$$= e^{-\lambda n\pi} (-e^{-\lambda\pi} + 1)$$

$$= e^{-\frac{n\pi}{\pi}} (1 - e^{-1})$$

$$\boxed{P_N(n) = e^{-n} (1 - e^{-1})} \quad \text{--- 2}$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

• Continuous random variables:

$$F_X(x) = p(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Home assignment

CLO No 4 - Construct probability distributions from given data for discrete and continuous random variables.

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part a (10 points): A Random Variable Z is defined as $Z = \min(X_1, X_2, \dots, X_n)$, where X_i 's are independent random variables with X_i having an exponential distribution with rate λ_i for $i = 1, \dots, n$.

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part a (10 points): A Random Variable Z is defined as $Z = \min(X_1, X_2, \dots, X_n)$, where X_i 's are independent random variables with X_i having an exponential distribution with rate λ_i for $i = 1, \dots, n$.

X_i are exponentially distributed, and so the PDF of each is:

$$f_{X_i}(x_i) = \begin{cases} \lambda_i e^{-\lambda_i x_i} & \text{if } x_i > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Now, we need to find out the CDF of Z . Since X_i cannot take values below 0, it is clear that $F_Z(z) = 0$ for all $z < 0$. For the case $z \geq 0$,

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\min(X_1, \dots, X_n) \leq z) \\ &= 1 - P(\min(X_1, \dots, X_n) > z) \\ &= 1 - P(X_1 > z, X_2 > z, \dots, X_n > z) \end{aligned}$$

Since X_i are independent,

$$F_Z(z) = 1 - P(X_1 > z) P(X_2 > z) \cdots P(X_n > z)$$

Let's find out the probability that $X_i > z$.

$$P(X_i > z) = \int_z^\infty \lambda_i e^{-\lambda_i x_i} dx_i = e^{-\lambda_i z}$$

Thus,

$$\begin{aligned} F_Z(z) &= 1 - e^{-\lambda_1 z} e^{-\lambda_2 z} \cdots e^{-\lambda_n z} \\ &= 1 - e^{-(\lambda_1 + \cdots + \lambda_n)z} \end{aligned}$$

Thus,

$$F_Z(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 - e^{-(\lambda_1 + \cdots + \lambda_n)z} & \text{if } z \geq 0. \end{cases}$$

Differentiating this, we can obtain PDF for Z :

$$f_Z(z) = \begin{cases} (\lambda_1 + \cdots + \lambda_n) e^{-(\lambda_1 + \cdots + \lambda_n)z} & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

The PDFs of two variables are:

$$f_X(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2} & \text{if } y \in [0, 2] \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\max(X, Y) \leq z) \\ &= P(X \leq z, Y \leq z) \end{aligned}$$

Problem 3 (20 points): Find the CDF and PDF for the Random Variable Z and **specify the range of values where they are applicable:**

Part b (10 points): Let X and Y be independent random variables that are uniformly distributed with $X \sim U[0, 1]$ and $Y \sim U[0, 2]$. The random variable $Z = \max(X, Y)$.

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$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\max(X, Y) \leq z) \\ &= P(X \leq z, Y \leq z) \end{aligned}$$

By independence of X and Y ,

$$\begin{aligned} &= P(X \leq z) P(Y \leq z) \\ &= \begin{cases} 0 & \text{if } z < 0 \\ z \times \frac{1}{2} & \text{if } 0 \leq z \leq 1 \\ 1 \times \frac{1}{2} & \text{if } 1 \leq z \leq 2 \\ 1 \times 1 & \text{if } z > 2 \end{cases} \end{aligned}$$

Congrats!

End of Unit 5!

اپنی ہستی ہی سے ہو جو کچھ ہو
آگہی گر نہیں غفلت ہی سہی

Unit 6!

Derived Distributions

A general function $g(X)$ of a continuous RV

1. Find the CDF of Y: $F_Y(y) = P(Y \leq y) = P(g(x) \leq y)$

2. Differentiate: $f_Y(y) = \frac{dF_Y}{dy}(y)$

Example 1: $Y = X^3$; X is uniform on $[0, 2]$

$F_Y(y) = P(Y \leq y)$ where $0 \leq y \leq 8$

$$= P(x^3 \leq y)$$

$$= P\left(x \leq y^{\frac{1}{3}}\right)$$

$$= \frac{1}{2} y^{1/3}$$

$$f_Y(y) = \frac{dF_Y}{dy}(y) = \frac{1}{2} \cdot \frac{1}{3} y^{-2/3}$$

$$= \frac{1}{6} \cdot \frac{1}{y^{2/3}}$$

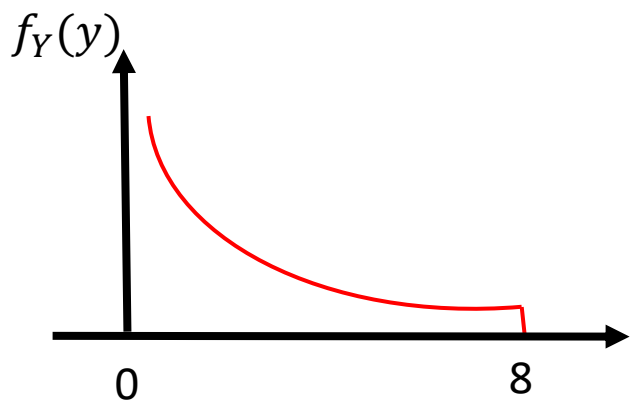
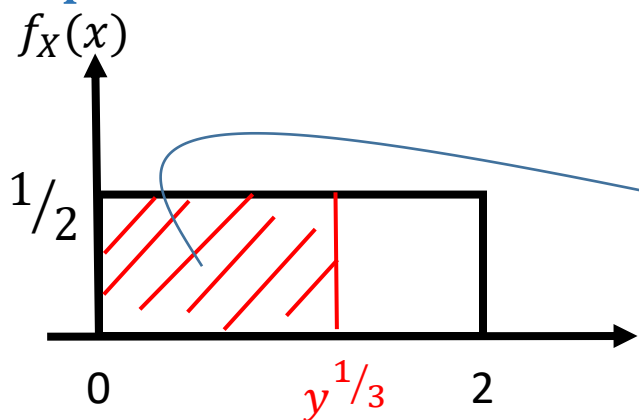


TABLE 5.1: AREA $\Phi(y)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF y

y	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

○ General normal $N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Quiz No 10

25:00

- 1) Present your final answers and reasoning clearly.
- 2) Most of the total score for the questions will be awarded on your reasoning.

TABLE 5.1: AREA $\Phi(y)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF y

y	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
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.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
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1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

○ General normal $N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Quiz No 10
(extra time)

05:00

- 1) Present your final answers and reasoning clearly.
- 2) Most of the total score for the questions will be awarded on your reasoning.



**Course & Instructor evaluations
– please!**



*Thank you &
Good Luck
The Last Lecture*

وہ بات سارے فسانے میں جس کا ذکر نہ تھا
وہ بات ان کو بہت ناگوار گزری ہے

